

ASHUTOSH ROY

Machine Learning Engineer | GenAI & Agentic AI

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SUMMARY

Machine Learning Engineer specializing in GenAI workflows, agentic AI components, and prompt engineering on cloud platforms. Expertise in building RAG pipelines, agent loops, planners, and tool-augmented workflows using LangGraph, MCP, and ADK frameworks. Published researcher with hands-on experience deploying production-ready AI systems via FastAPI with clean code, unit testing, observability, and Responsible AI principles.

EXPERIENCE

Research Intern

Jan 2025 – Jun 2025

Indian Institute of Technology (IIT) Delhi

New Delhi, India

- Implemented advanced GenAI workflows including embeddings and RAG pipelines (FAISS/Pinecone + Apache Solr), optimizing semantic search and agent tool usage over multimodal archaeological datasets.
- Deployed scalable **FastAPI** services for real-time LLM inference with safety prompts, IAM-based access controls, and robust error handling; reduced RAG latency by 35%.
- Applied Responsible AI principles to document understanding pipelines (Gemini OCR); contributed to team documentation, knowledge-sharing wikis, and design discussions.
- Co-authored **SARCH: Multimodal Search for Archaeological Archives** — submitted for peer review. arxiv.org/abs/2511.05667

AI/ML Engineer Intern

Oct 2024 – Nov 2024

MitoVoid AI

Remote

- Built end-to-end Python ML pipelines and agent tools for healthcare, creating conversational AI agents using prompt engineering, memory fundamentals, and autonomous tool-calling capabilities.
- Deployed a FastAPI chatbot backend with robust error handling, retries, and observability (logs); reduced patient query resolution time by ~40% through optimized intent routing.
- Collaborated closely with senior engineers on architecture design and code reviews, adopting best practices for model quantization and unit testing to optimize inference latency by 30%.

PROJECTS

Shopping Assistant Agent (LangGraph & MCP) | Python, LangGraph, MCP, Claude 3.5, FastAPI

Feb 2026 - May 2026

- Engineered a multi-agent shopping assistant using **LangGraph** to coordinate specialized agents for product search, comparison, recommendation, and review summarization with self-reflection and retry loops.
- Implemented custom **Model Context Protocol (MCP)** servers enabling secure real-time access to product APIs, shopping databases, pricing tools, and external e-commerce sources for contextual recommendations.

Skills for Agents Framework | Open-source, Python, AI-Agents

April 2026 - May 2026

- Developed a production-ready ecosystem of composable instruction sets, allowing AI agents to handle multiple tasks concurrently without prompt collision.
- Architected the **Skills Interoperability Protocol (SIP)**, dividing capabilities into domains (voice, density, craft) enabling complex layered and sequential pipeline agent behaviors.

Exam-Helper Agentic RAG | Python, LLaMA 3, LangGraph, FAISS, Pinecone

Dec 2024 - Apr 2025

- Built an LLM-powered autonomous contextual QA agent on **LLaMA 3** using LangGraph for multi-step reasoning across PDFs, images, and YouTube transcripts.
- Implemented dense retrieval (FAISS) with semantic chunking and dynamic context augmentation to drastically reduce hallucination in domain-specific academic queries.

Speech Emotion & Stress Detection | PyTorch, Wav2Vec2, BiLSTM, Hugging Face | RECCAP 2026

Aug 2024 - Dec 2024

- Proposed a self-supervised multitask architecture for simultaneous emotion classification and stress regression; **accepted at RECCAP 2026, IIT Palakkad.**

Medical Chatbot (LLM Fine-Tuning) | Python, Mistral 7B, QLoRA, Tesseract OCR

Mar 2024 - Apr 2024

- Fine-tuned **Mistral 7B** using **QLoRA** for domain-specific medical NLU, optimizing inference speed while preserving high intent-recognition accuracy.

TECHNICAL SKILLS

GenAI & Agentic AI: Prompt Engineering, Embeddings, RAG Pipelines, Agent Loops, Planners, Memory, Tool Integration (ADK, LangGraph, MCP)

Cloud & GCP: Vertex AI, BigQuery, Cloud Storage, Pub/Sub, Cloud Run, IAM Roles, Secret Management

Coding & Backend: Python (Proficient), FastAPI, REST APIs, SDKs, Unit Testing, Observability (Logs/Errors/Retries), Docker

Databases & ML: FAISS, Pinecone, Apache Solr, Transformer Fine-Tuning (QLoRA), SQL, Git, GitHub Actions

EDUCATION

Chhattisgarh Swami Vivekananda Technical University (CSVTU)

2022 – 2026

B.Tech (Hons) in Computer Science and Engineering (Artificial Intelligence)

Bhilai, Chhattisgarh

St. Columbus School

2020 – 2021

Higher Secondary (Class XII - Science PCMB) - 87%

Faridabad, Haryana

St. Columbus School

2018 – 2019

Secondary (Class X)

Faridabad, Haryana